

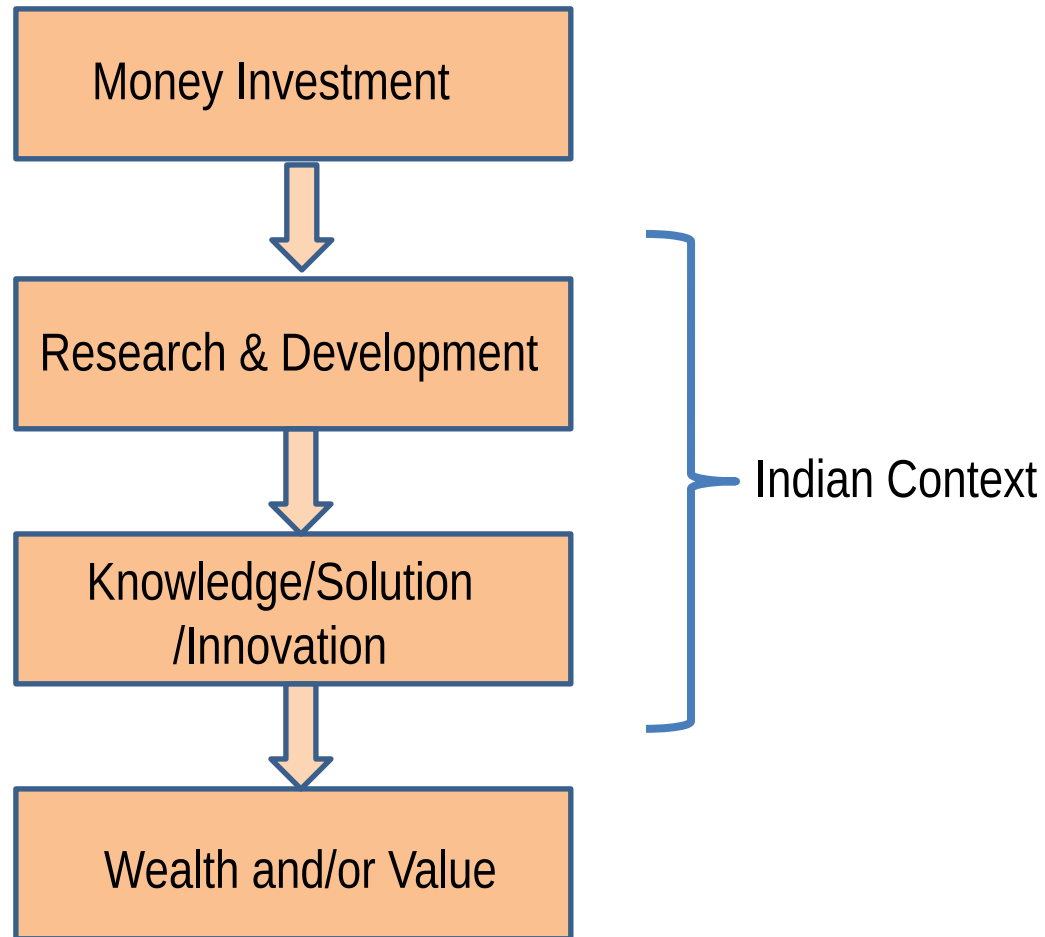
Academic Writing and Quality of Publications

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Identification of Research Question

- The research question chosen for any scientific research is expected to be of Indian context. Indian funding agencies (Govt or private) are interested to sponsor R&D research, if the problem chosen is of Indian context.
- Money received from any Funding Agency to carry out scientific research on a live research problem are invested in
 - manpower,
 - procurement of equipment,
 - travel, consumables and contingenciesto achieve target objectives as per schedule.
- Once research outcomes are visible, technical writing in an appropriate way is essential for its dissemination.
- Research outcomes are expected to be novel, significant and able to advance the domain knowledge.

Research, Development & Innovation



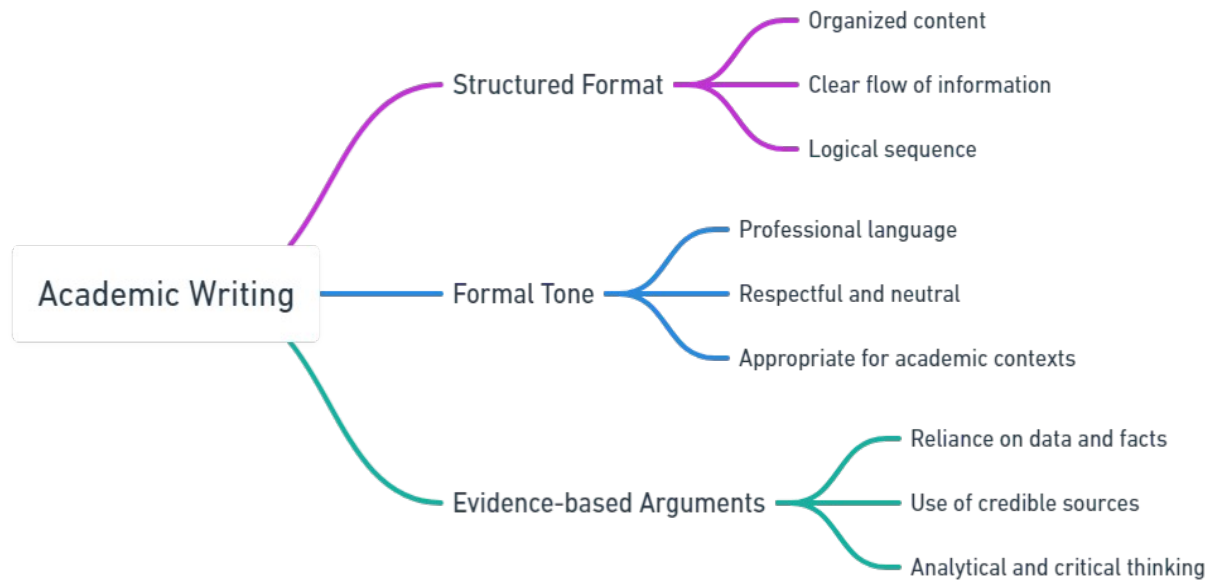
Research based solutions successfully deployed for handling socio-economic problems.
Paradigm of Innovation is usually country and context specific.

Academic Writing: Technical Proposals / Papers

- Research Proposal Writing
 - Grants from local bodies
 - Grants from National bodies
 - Grants from Int’nl bodies
- Technical Article Writing
 - Conference paper/ Journal paper / Technical Reports
- Patent Application Writing
 - National / International

Academic Writing

Academic writing is a structured and formal way of communicating your research and ideas.



Importance of Academic Style

- **Reflects Your Understanding**: Demonstrates your grasp of the subject matter.
- **Critical Thinking**: Shows your ability to analyze, evaluate, and synthesize information.
- **Credibility**: Builds trust in your arguments by adhering to academic standards.
- **Professionalism**: Conveys your ideas in a structured and polished manner.
- **Audience Engagement**: Helps communicate complex ideas clearly to peers and academics.

Characteristics of Academic Writing



Accuracy: Ensure your information is correct and well-supported.



Brevity: Be concise; stay focused on the topic.



Clarity: Make your ideas easy to follow.



Formality: Use a formal tone and avoid casual language.



Consistency: Maintain a uniform style and voice throughout your writing.

Strategies for Effective Writing



Be Relevant: Directly address the essay or research question to stay on topic and meet the assignment's objectives.



Organize Your Thoughts: Use clear structure and logical flow to guide the reader through your arguments seamlessly.



Follow Academic Conventions: Ensure proper citations and referencing to give credit to sources and avoid plagiarism.



Use Evidence-Based Arguments: Support your claims with data, research findings, and credible sources to strengthen your case.

Continuous Improvement in Academic Writing

- **Writing as a Process**

- Practice and Feedback: Regular practice and constructive feedback are essential for refining your writing skills.
- Revisit and Revise: Regularly reviewing and revising your work helps you identify areas for improvement.

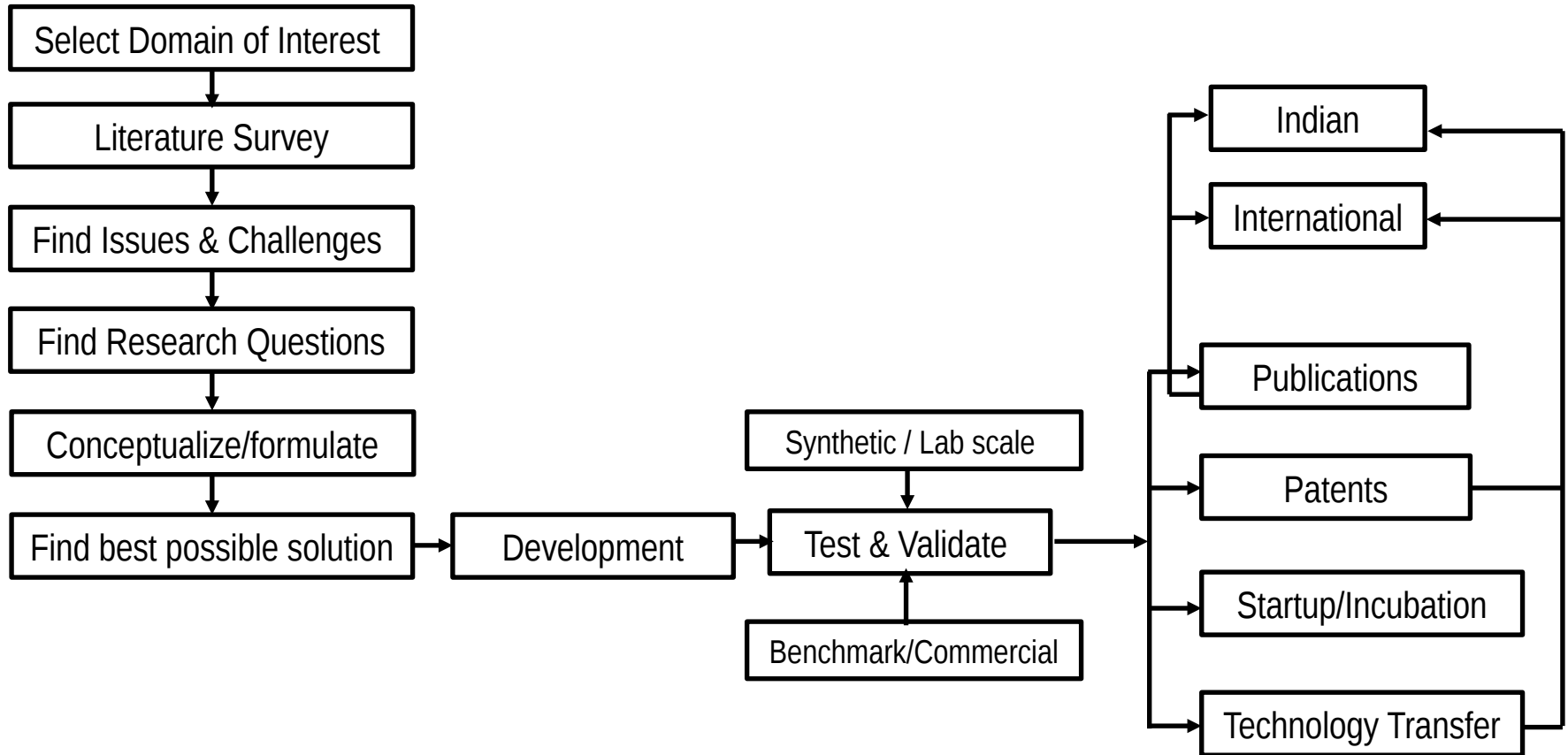
- Use Feedback Constructively**

- Implement Suggestions: Apply the feedback you receive to enhance your writing in future assignments.
- Reflect on Mistakes: Understand the reasons behind any errors and learn from them to avoid repeating them.

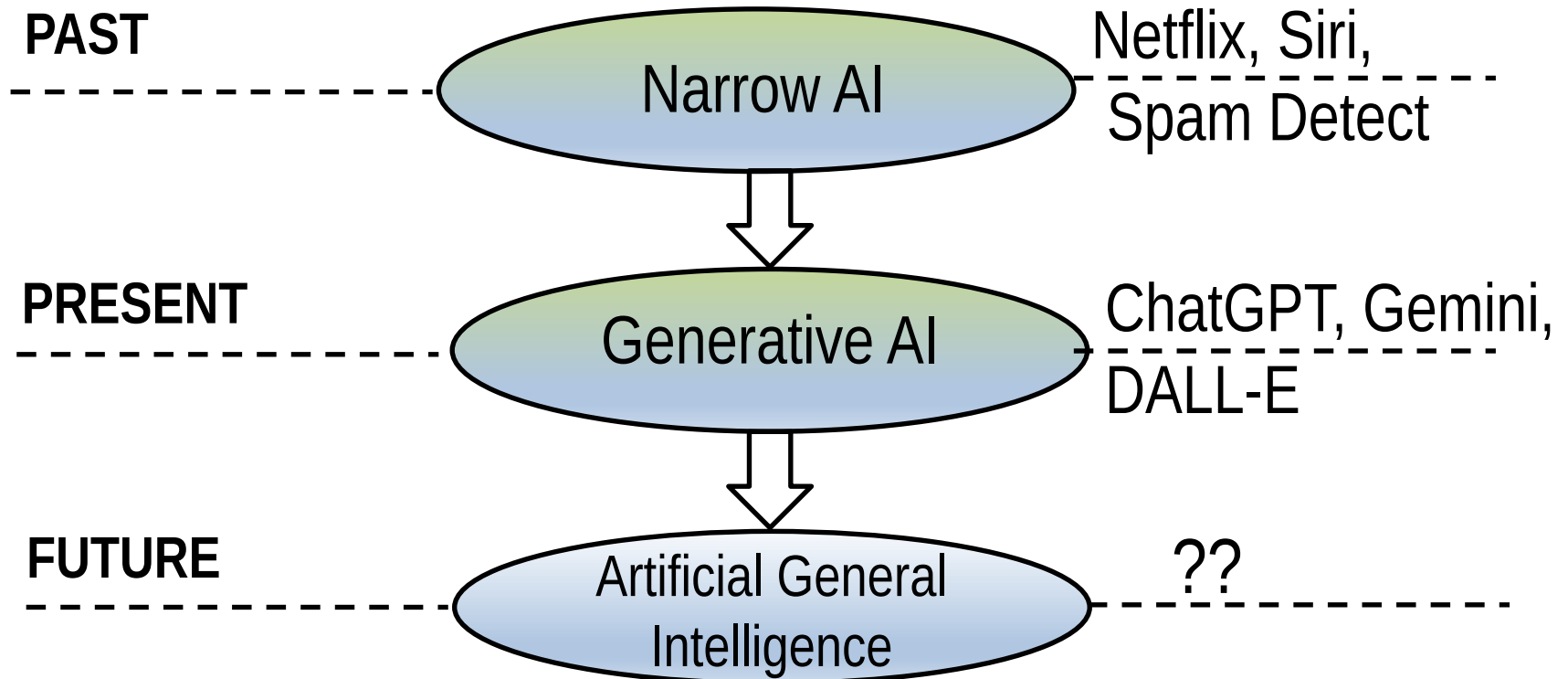
- Engage with a Variety of Texts:**

- Read Widely: Exposure to different writing styles and academic works can improve your own writing.
- Analyze Structure and Language: Pay attention to how arguments are constructed and how language is used in other academic texts.

Research Article Publication Process



AI in Literature Survey



- **ChatGPT** and **Gemini** are largely responsible for the considerable buzz around **Generative AI**, which uses data from **ML** models to **answer questions and create images, text and videos**.
- Large language models (LLM) are very large deep learning models that are pre-trained on vast amounts of data.

Generative AI and ChatGPT

- **ChatGPT** is a specific implementation of **Generative AI** (a broad field of AI) that helps with content creation and information retrieval.
- ChatGPT is an AI-based service that you can access via the Internet. One can use it to organize or summarize text, or to write new text.
- It enables to understand and respond to user questions and instructions. It “reads” a large amount of text and learns how words tend to appear in context with other words.
- It uses what it has learned **lawfully** using **large language model** and can generate text on a seemingly endless range of topics.
- LLMs use a particular neural network architecture called a transformer, which is designed to process and generate data in sequence, like text.
- **OpenAI's** ChatGPT (introduced in December'22) uses **three primary sources** of information:
 - publicly available information on the internet,
 - information that we license from third parties, and
 - information that is provided by users or human trainers.

Generative AI Tools: Academic vs General

Chatbot

ChatGPT

Poe (ChatGPT,
GPT-4, Claude)

Chatbot+ Academic

Scite Assistant

Perplexity

Chatbot+

Perplexity

Poe (Web search)

Academic

Elicit

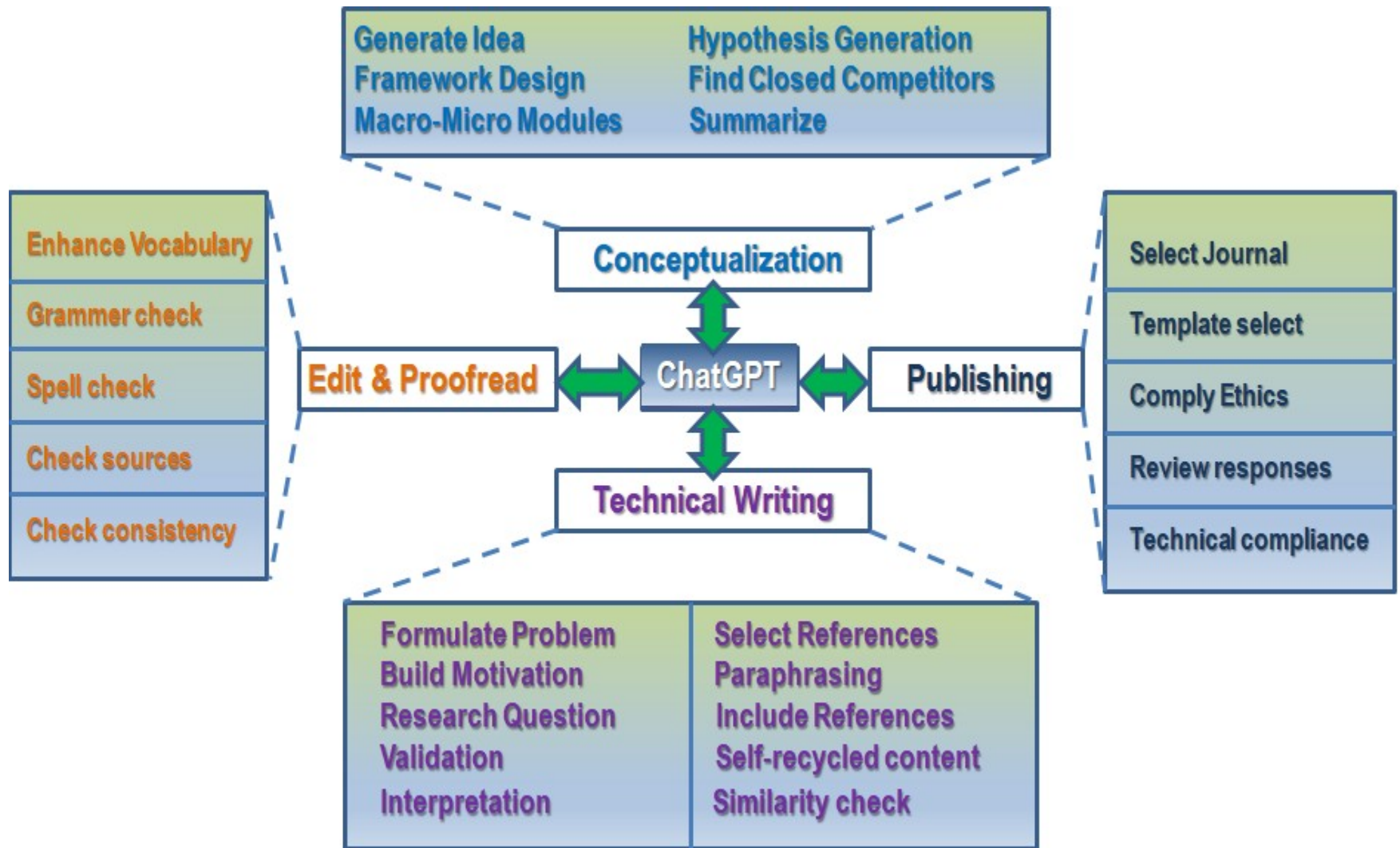
Consensus

Note-taking

Notion AI

Chatbot: General Question-answer
Search content: Get updated infor.
Academic: Get scholarly contents

Ethical Use of Generative AI in Research



Common AI Tools for Literature Survey

- Scite
 - Scite is a smart citation tool that analyzes and classifies citation context to determine whether an article provides supporting or contrasting evidence. The tool helps researchers and students easily find relevant and well-supported results, track and improve research outcomes, and discover and evaluate scientific articles.
- SciSpace by Typeset
 - Typeset is an authoring and collaboration platform for researchers. It provides an editor designed for research writing, with features like automated formatting, reference management, and collaboration tools. It also offers a library of templates and tools to help researchers write and submit better papers , as well as for publishers to generate XML, PDF, HTML, and ePUB in a single click.

Common AI Tools for Literature Survey

- Perplexity AI

- Perplexity AI chat tool aimed at acting as an extremely powerful search engine . The tool uses large language models to answer questions posed by users. It is an answer engine that aims to deliver accurate answers to questions using AI.

- ResearchRabbit

- Researcher rabbit is a platform build for researchers, it provides a lot of tools to help discover, visualize and gain insights from research papers. In addition to that it helps researchers collaborate and comment on research papers among other features.

Common AI Tools for Literature Survey

- Connected Papers

- A unique, visual tool to help researchers and applied scientists find and explore papers relevant to their field of work. A research and knowledge-sharing platform that helps scientists, researchers, and students find the latest research and insights about their topics of interest.

- Elicit

- Elicit is an AI research assistant that uses language modeling to automate research workflows, including literature reviews. The tool can find relevant papers without perfect keyword matching, summarize takeaways, and extract key information from papers. In addition to answering research questions, Elicit can also help with brainstorming, summarizing, and text classification.

Common AI Tools for Literature Survey

- Scholarcy
 - Scholarcy is an online summarizing tool that helps people to quickly summarize long pieces of text. It uses AI techniques to extract the main points and key information from long documents, allowing users to gain an understanding of the text in minutes.
- HeyScience
 - Heyscience's Person AI Assist is a tool that can analyze millions of scientific research papers quickly, making the time-consuming, tedious task much easier for researchers. It helps users stay up-to-date with research trends and related documents. The tool can also provide short, simplified summaries of scientific articles and provide pre-submission reviews with feedback. Users can join the waitlist using their email address for early access.

Common AI Tools for Literature Survey

- Semanticscholar
 - The Semant Scholar AI-powered research tool is a free scientific literature search with over 211 million papers in the field of science. It also offers an API for developers to build Scholar apps and a beta version of Semant Reader, which augments scientific reading. The tool is based on the Allen Institute AI and aims to increase efficiency and inclusivity in AI research while decreasing carbon footprint.
- Sourcely
 - Sourcely is an AI-powered tool designed to help students and researchers discover reliable sources effortlessly. It offers features like source finding based on essay titles or content, summarizing articles of interest, and filtering results by publication year and citations. With Sourcely, users can save time on research and academic writing by simply pasting their essay or providing the title, allowing the tool to generate suitable sources.

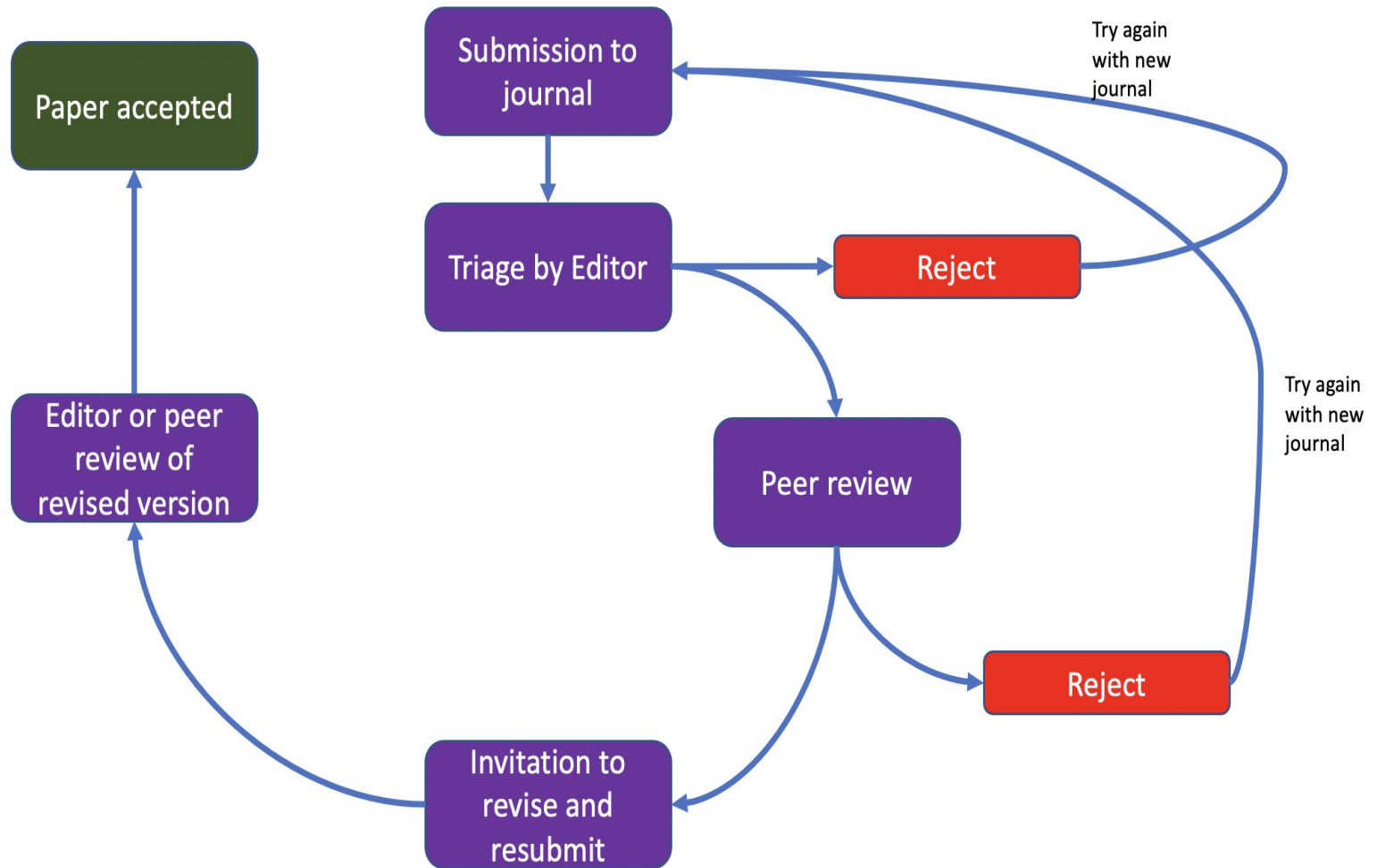
Common AI Tools for Literature Survey

- Lumina Chat
 - Lumina is an AI-powered research suite that allows users to access a vast database of 300k journal articles. It provides accurate and cited answers to research queries and enables users to make collections of relevant sources. Lumina offers dedicated bots to carry out research tasks and is supported by most web browsers. The main features of Lumina include the ability to ask questions and receive cited answers from open-source articles, dig deeper into a topic to test hypotheses, and explore contextual information.
- Mirrorthink
 - The Mirrorthink AI tool is a powerful resource for enhancing scientific research. It allows users to explore and analyze existing academic work, gaining a comprehensive understanding of a specific topic or research question. Users can also perform mathematical calculations and understand the execution logic behind them. Additionally, Mirrorthink offers valuable insights into the scientific market, helping users to discover trends and create market segmentations.

Research Publication Steps

- Once a goal/objective has been achieved, the researcher may plan to publish the work as peer-reviewed
 - Int’nl/ National Journal (Tier I or II) article,
 - Int’nl/ National Conference (Scopus or high h5 rank) article, or
 - Int’nl or National Book chapter.
- After submission, there may be four possible decisions-
 - Accepted,
 - Major/Minor revision, or
 - Reject.
- Revisions or rejection should be taken positively and attempt sincerely to improve the standard of the MS. After all we are now part of a professional knowledge society.
- Publishing work in quality journal/proceedings/books help improve visibility of the researcher across the globe.

Research Publication Steps



What is Citation?

A "citation" is the way you tell your readers that certain material in your work came from another source. It also gives your readers the information necessary to find that source again, including:

- information about the author
- the title of the work
- the name and location of the company that published your copy of the source
- the date your copy was published
- the page numbers of the material you are borrowing

Why should I cite sources?

Giving credit to the original author by citing sources is the only way to use other people's work without plagiarizing. But there are a number of other reasons to cite sources:

- citations are extremely helpful to anyone who wants to find out more about your ideas and where they came from
- not all sources are good or right -- your own ideas may often be more accurate or interesting than those of your sources. Proper citation will keep you from taking the rap for someone else's bad ideas
- citing sources shows the amount of research you've done
- citing sources strengthens your work by lending outside support to your ideas

Doesn't citing sources make my work seem less original?

Not at all. On the contrary, citing sources actually helps your reader distinguish your ideas from those of your sources. This will actually emphasize the originality of your own work.

When do I need to cite?

Whenever you borrow words or ideas, you need to acknowledge their source. The following situations almost always require citation:

- whenever you use quotes
- whenever you paraphrase
- whenever you use an idea that someone else has already expressed
- whenever you make specific reference to the work of another
- whenever someone else's work has been critical in developing your own ideas.

How do I cite sources?

This depends on what type of work you are writing, how you are using the borrowed material, and the expectations of your instructor.

1. You have to think about how you want to identify your sources. If your sources are very important to your ideas, you should mention the author and work in a sentence that introduces your citation. If, however, you are only citing the source to make a minor point, you may consider using parenthetical references, footnotes, or endnotes.
2. There are also different forms of citation for different disciplines. For example, when you cite sources in a psychology paper you would probably use a different form of citation than you might in a paper for an English class.
3. You should always consult your instructor to determine the form of citation appropriate for your paper. You can save a lot of time and energy simply by asking "How should I cite my sources," or "What style of citation should I use?" before you begin writing.

Identifying sources in the Body?

The first time you cite a source, it is almost always a good idea to mention its author(s), title, and genre (book, article, or web page, etc.). If the source is central to your work, you may want to introduce it in a separate sentence or two, summarizing its importance and main ideas. But often you can just tag this information onto the beginning or end of a sentence. For example, the following sentence puts information about the author and work before the quotation:

Milan Kundera, in his book The Art of the Novel, suggests that “if the novel should really disappear, it will do so not because it has exhausted its powers but because it exists in a world grown alien to it.”

Identifying sources in the Body? (contd.)

You may also want to describe the author(s) if they are not famous, or if you have reason to believe your reader does not know them. You should say whether they are economic analysts, artists, physicists, etc. If you do not know anything about the author, and cannot find any information, it is best to say where you found the source and why you believe it is credible and worth citing. For example,

In an essay presented at an Asian Studies conference held at Duke University, Sheldon Geron analyzes the relation of state, labor-unions, and small businesses in Japan between 1950s and 1980s.

Identifying sources in the Body? (contd.)

If you have already introduced the author and work from which you are citing, and you are obviously referring to the same work, you probably don't need to mention them again. However, if you have cited other sources and then go back to one you had cited earlier, it is a good idea to mention at least the author's name again (and the work if you have referred to more than one by this author) to avoid confusion.

Quoting Material

What is quoting?

Taking the exact words from an original source is called quoting. You should quote material when you believe the way the original author expresses an idea is the most effective means of communicating the point you want to make. If you want to borrow an idea from an author, but do not need his or her exact words, you should try paraphrasing instead of quoting.

Quoting Material

(contd.)

How often should I quote?

Most of the time, paraphrasing and summarizing your sources is sufficient (but remember that you still have to cite them!). If you think it's important to quote something, an excellent rule of thumb is that for every line you quote, you should have at least two lines analyzing it.

Quoting Material

(contd.)

How do I incorporate quotations in my paper?

Most of the time, you can just identify a source and quote from it, as in the first example above. Sometimes, however, you will need to modify the words or format of the quotation in order to fit in your paper. Whenever you change the original words of your source, you must indicate that you have done so. Otherwise, you would be claiming the original author used words that he or she did not use. But be careful not to change too many words! You could accidentally change the meaning of the quotation and falsely claim the author said something they did not.

Quoting Material

(contd.)

Quoting Within Quotes

When you have "embedded quotes," or quotations within quotations, you should switch from the normal quotation marks (""") to single quotation marks (") to show the difference. For example, if an original passage by John Archer reads:

The Mountain Coyote has been described as a “wily” and “single-minded” predator by zoologist Lma Warner.

Your quotation might look like this:

As John Archer explains, “The Mountain Coyote has been described as a 'wily' and 'single-minded' predator by zoologist Lma Warner.”

Quoting Material

(contd.)

How do I include long quotes in my paper?

Formatting requirements for long quotations differ depending on the citation style. If you are quoting more than 3 lines of material, you should do the following:

- change the font to smaller (say, 10 point font)
- double indent the quotation -- that means adjusting the left and right margins so that they are about one inch smaller than the main body of your paper
- "Left-justify" the text, so that each line begins in the same place, creating a straight line on the left side of the quotation, while the right side is jagged
- Do NOT use quotation marks for the entire quotation -- the graphic changes you have made already (changing the font, double indenting, etc.) are enough to indicate that the material is quoted. For quotations within that quotation, use normal quotation marks, not single ones
- Skip 1.5 times the line-spacing you are using in the document before you begin the quotation and after it. This is optional and depends on the style preferred by your instructor

Listing References

What's a Bibliography?

A bibliography is a list of all of the sources you have used in the process of researching your work. In general, a bibliography should include:

- the authors' names
- the titles of the works
- the names and locations of the companies that published your copies of the sources
- the dates your copies were published
- the page numbers of your sources (if they are part of multi-source volumes)

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Listing References

(contd.)

What's an Annotated Bibliography?

An annotated bibliography is the same as a bibliography with one important difference: in an annotated bibliography, the bibliographic information is followed by a brief description of the content, quality, and usefulness of the source.

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Listing References

(contd.)

What are Footnotes?

- Footnotes are notes placed at the bottom of a page. They cite references or comment on a designated part of the text above it. For example, say you want to add an interesting comment to a sentence you have written, but the comment is not directly related to the argument of your paragraph. In this case, you could add the symbol for a footnote. Then, at the bottom of the page you could reprint the symbol and insert your comment.
- This is an illustration of a footnote.¹ The number “1” at the end of the previous sentence corresponds with the note below.

¹ At the bottom of the page you can insert your comments about the sentence preceding the footnote.

Listing References

(contd.)

What's the difference between Footnotes and Endnotes?

The only real difference is placement -- footnotes appear at the bottom of the relevant page, while endnotes all appear at the end of your document. If you want your reader to read your notes right away, footnotes are more likely to get your reader's attention. Endnotes, on the other hand, are less intrusive and will not interrupt the flow of your paper.

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Listing References

(contd.)

If I cite sources in the Footnotes (or Endnotes), how's that different from a Bibliography?

- Sometimes you may be asked to include these -- especially if you have used a parenthetical style of citation. A "works cited" page is a list of all the works from which you have borrowed material. Your reader may find this more convenient than footnotes or endnotes because he or she will not have to wade through all of the comments and other information in order to see the sources from which you drew your material. A "works consulted" page is a complement to a "works cited" page, listing all of the works you used, whether they were useful or not.

Citation Styles

Citation styles differ mostly in the location, order, and syntax of information about references. The number and diversity of citation styles reflect different priorities with respect to concision, readability, dates, authors, publications, and, of course, style.

There are also two major divisions within most citation styles: documentary-note style and parenthetical style. Documentary-note style is the standard form of documenting sources. It involves using either footnotes or endnotes, so that information about your sources is readily available to your readers but does not interfere with their reading of your work. .

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Citation Styles: Humanities

Chicago

- [Writer's Handbook: Chicago Style Documentation](#)
- [Excellent FAQ on Usage in the Chicago Style](#)
- [Writer's Handbook: Chicago Style Documentation](#)

MLA (Modern Language Association)

- [Writer's Handbook: MLA Style Documentation](#)
- [MLA Citation Style](#)

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Citation Styles: Science & Engg

ACS (American Chemical Society)

- [AMA Citation Style](#)

IEEE (Institute of Electrical and Electronics Engineers)

- [Electrical Engineering Citation Style](#)

NLM (National Library of Medicine)

- [NLM Style Guide](#)
- [National Library of Medicine Recommended Formats for Bibliographic Citation \(PDF format\)](#)

Vancouver (Biological Sciences)

- [Introduction to the Vancouver Style](#)

Citation Styles: Social Sciences

AAA (American Anthropological Association)

- [Citations and Bibliographic Style for Anthropology Papers](#)

APA (American Psychological Association)

- [Writer's Handbook: APA Style Documentation](#)
- [APA Style.org](#)

APSA (American Political Science Association)

- [Writer's Handbook: APSA Documentation](#)

Legal Style

- [Legal Citation: Using and Understanding Legal Abbreviations](#)
- [Legal Research and Citation Style in the USA](#)

How to Paraphrase?

To paraphrase is to include the ideas or information from an original source in your paper by rephrasing those ideas or information in *your own words*. The key to successful paraphrasing is to use as few words as possible from the original text--be mindful not to change the meaning that you are trying to convey as you rephrase--and to cite your paraphrase.

Without proper citation, your paraphrase could be construed as plagiarism.

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Impact Factor of a Research Journal

In a given year, it is the average number of citations received per paper published in that journal during the two preceding years.

$$\text{Impact Factor (2024)} = \frac{\text{Citations in 2024 to articles published in 2023 \& 2022}}{\text{Articles published in 2023 and 2022}}$$

Importance of Research Publication in Quality Journals

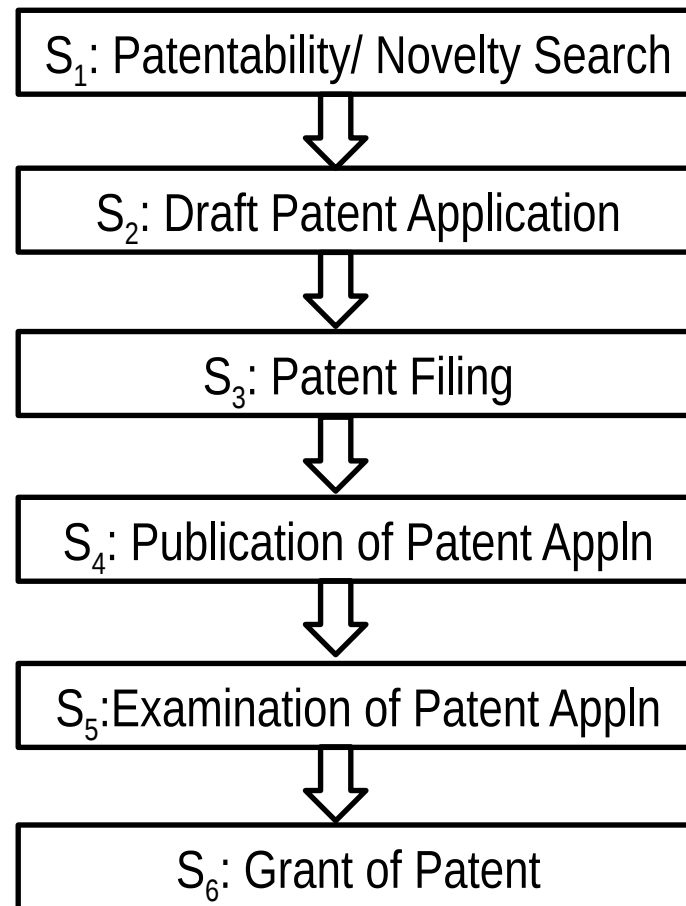


Patent Registration in India: Steps

- A patent is a public recognition for an invention.
- A patent refers to a unique solution for a given problem with sufficient information about the solution that it can be practicable by others.
- To obtain a patent, patent processing includes six (S_1 - S_6) steps and it must be systematically documented, validated, and defended.
- An invention is considered patentable, if it is
 - statutory,
 - novel,
 - useful, and
 - non-obvious.

To establish an invention to be novel and non-obvious, it is often essential to carry out an appropriate patent literature search with the assistance of a professional or an attorney or agent.

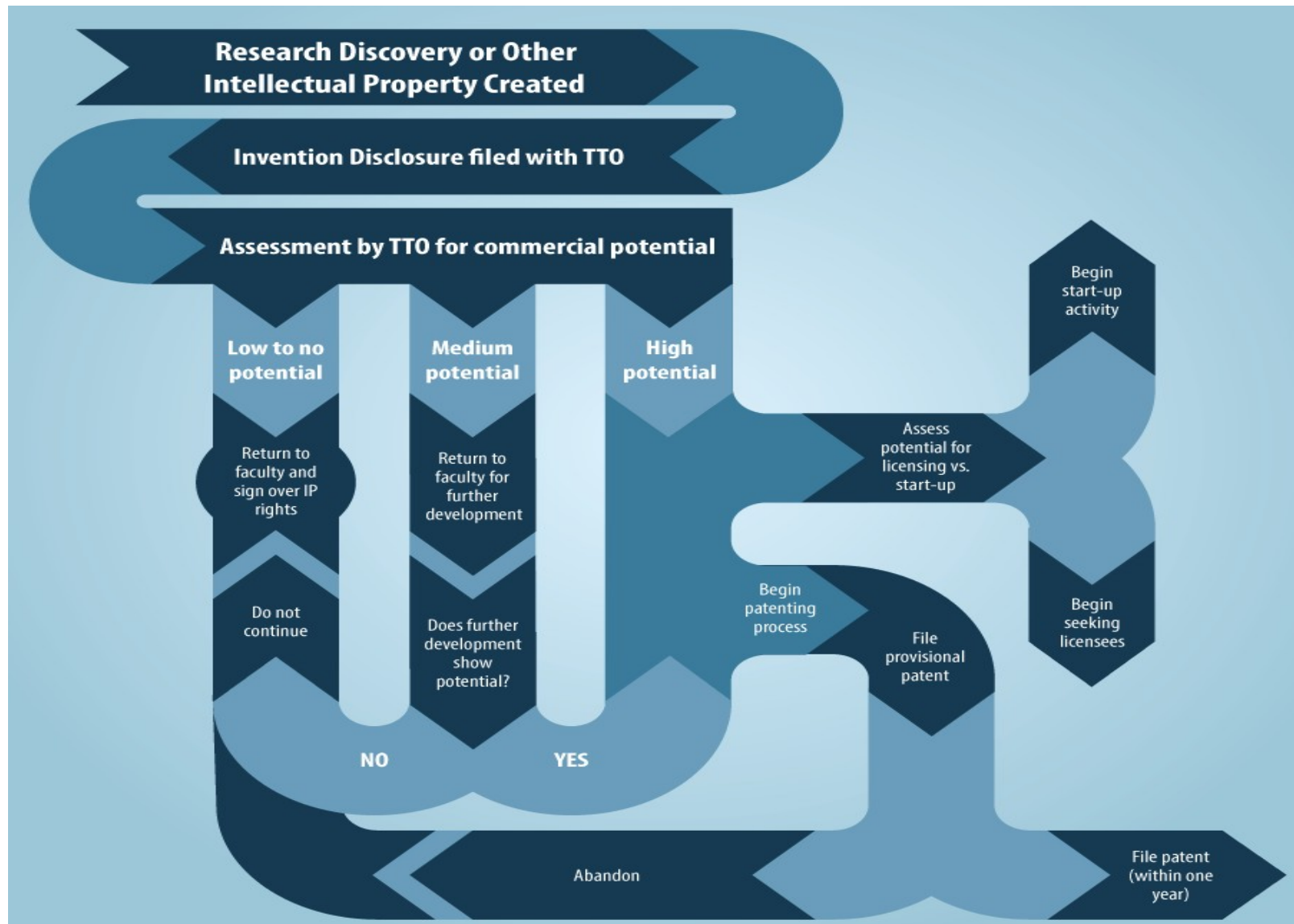
Patent Registration in India: Steps



Technology Transfer & Commercialization

- Technology Transfer and Commercialization involves Science, Law, and Business activities that lead to the development of products beneficial (directly or indirectly) for the society.
- There are four overall stages in the technology transfer/commercialization process:
 - Invention Disclosure,
 - Protection,
 - Marketing, and
 - Licensing

Technology Transfer & Commercialization



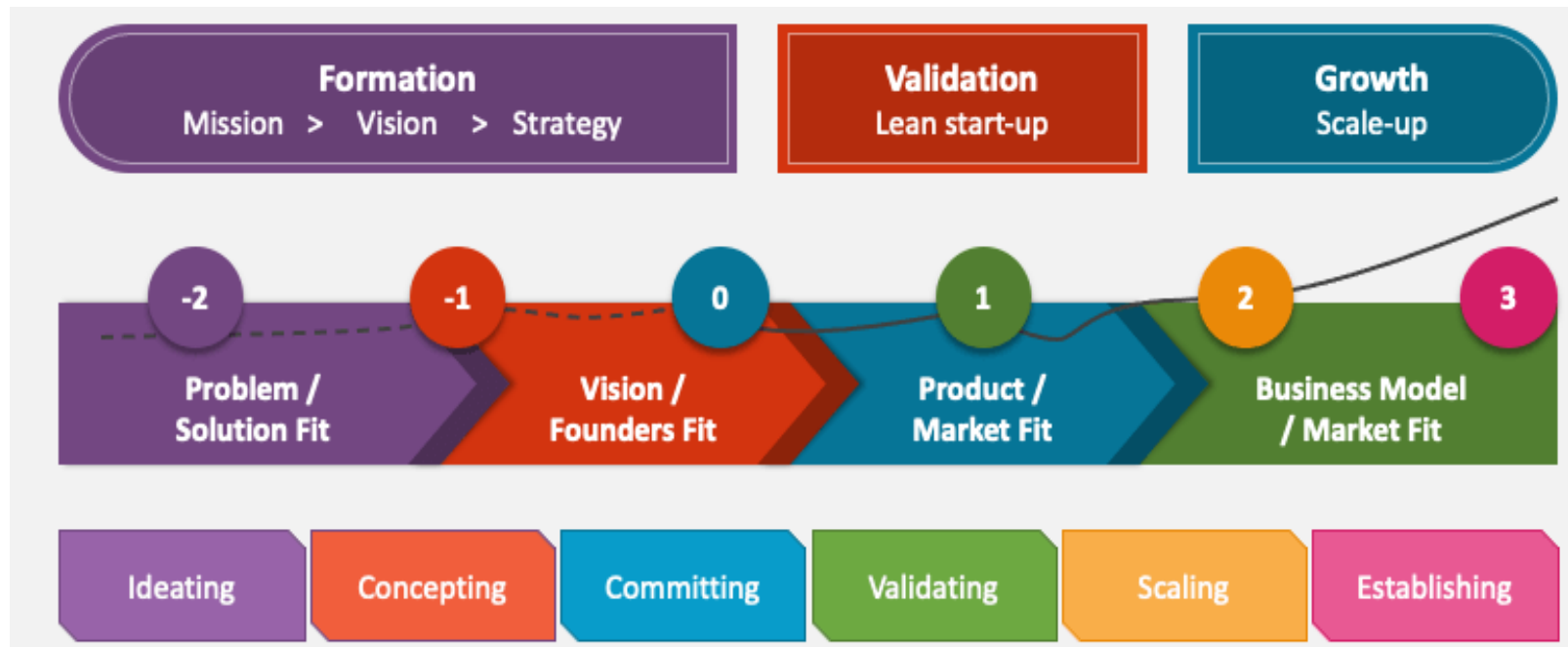
Technology Readiness Levels

RESEARCH DEVELOPMENT DEPLOYMENT	9	ACTUAL SYSTEM PROVEN IN OPERATIONAL ENVIRONMENT
	8	SYSTEM COMPLETE AND QUALIFIED
	7	SYSTEM PROTOTYPE DEMONSTRATION IN OPERATIONAL ENVIRONMENT
	6	TECHNOLOGY DEMONSTRATED IN RELEVANT ENVIRONMENT
	5	TECHNOLOGY VALIDATED IN RELEVANT ENVIRONMENT
	4	TECHNOLOGY VALIDATED IN LAB
	3	EXPERIMENTAL PROOF OF CONCEPT
	2	TECHNOLOGY CONCEPT FORMULATED
	1	BASIC PRINCIPLES OBSERVED

Startup Development Cycle

- The Startup Ecosystem of our country has been growing rapidly.
- The number of Startups increasing over 68 thousand between 2014 and June 2023, which has played a key role in catalyzing the Nation's Reputation as the hotbed of
 - Innovation,
 - Revolutionary ideas, and
 - Technologies.
- Country's Startup Scheme is based on three pillars:
 - To provide funding support and incentives to the start-ups.
 - To provide Academia-Industry Partnership and Incubation.
 - Simplification and Handholding.

Startup Development Cycle



Some HEI Initiatives

1. Expanding and deepening university-industry R&D.

Universities are generally generators of new knowledge/ideas, whereas industry is champion and shepherd of the process that enables the translation of ideas, knowledge and technologies, to successful application, providing society with useful products and services. Collaborative efforts can strengthen both.

- Supporting high-potential university-industry partnerships (Project-centered program)
- Industry-academia fellows program (Both way beneficial)
- Industry-academia Doctoral Fellowship program (Fellowship for those working on a industry/societal question)
- “Solution Partners” program (Fund for enterprises which aim to solve a problem with universities/R&D institutions)

Some Initiatives

2. Strategically strengthening high opportunity technology areas

Almost all major economies focus on high-opportunity technology areas since these can play a critical role industrial and economic development.

- Overseas Indian Partnership for High Impact

(India has large no of successful academic and entrepreneurial Diaspora who are generally open to contribute to their country)

- Creating Multi-Institutional Research and Technology Centres

Possible functions of MIRTC are: (i) targeted support for collaborative R&D, (ii) incentives and resources for participant institutions to deepen their interactions, (iii) shared critical research infrastructure, and (iv) cutting-edge technological knowledge.

Some Initiatives

3. Supporting and Stimulating development of solutions for high priority areas

- Transformational Technology R&D program.

(Rather than funding open-ended scientific research, it aims to fund for more targeted research to develop technologies that have the potential to have a major impact in the high priority areas in long run)

- Achieving Outcomes through Incentives

(It aims to use incentives as a way to promote the development of technologies where the pathway to the desired technology is not obvious)

- Global Technology Sandbox

(Govt will provide fund to startups or established enterprises to demonstrate their solutions, which will also allow them to refine their solutions for better fitting into the local needs)

Some Initiatives

4. Strengthening Technology Translation

(Recently, entrepreneurship for translation of technologies from ideas to application has been receiving much attention thru Atal Innovation Mission.)

- Meta incubator for technology entrepreneurship support

(Forming a large network of mentors across the country to advise selected technology-centered startups. Aims to set up (i) an “Expert Advice Vouchers” to reach the mentors and (ii) online technical advisory service.)

- Facilitating student engagement

(Fellowship program for engagement of students with the technology-translation process during their stay in a university, thereby helping to grow professionalism among them.)

Some Initiatives

5. Promote gender inclusion in Engineering & Technology Entrepreneurship

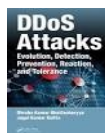
To enhance the number of women in Engineering and Technology entrepreneurship country-wide, whereas women constitute half of the population. A country with half of its talent pool not contributing towards development of the nation is definitely not desirable! It's a global problem and no single "silver bullet" solution. So, it demands a multi-dimensional approach to provide motivation, means, and opportunity for women to engage in research and development from an early stage onwards.

Concluding Remarks

- The importance of academic writing has been discussed.
- Characteristics and strategies to be adopted for quality academic writing has been presented.
- Importance of Research Paper publication has been discussed.
- A good number of AI based tools for literature survey has been discussed.
- Patent registration process has been discussed.
- Also, technology transfer and incubation process have been elaborated.
- Some Initiatives for HEI have been reported.



[Network Anomaly Detection: A Machine Learning Perspective](#), authored by D K Bhattacharyya and J K Kalita, published by CRC Press, 2013.



[DDoS Attacks: Evolution, Detection, Prevention and Reaction](#), authored by D K Bhattacharyya and J K Kalita, published by CRC Press, 2016.



[Network Traffic Anomaly Detection and Prevention](#), authored by D K Bhattacharyya, M H Bhuyan, Jugal K Kalita; published by Springer, 2017.



[Gene Expression Data Analysis: A Statistical and Machine Learning Perspective](#), authored by D K Bhattacharyya, P Barah and J K Kalita, published by CRC Press, 2021.



[Fundamentals of Data Science: Theory and Practice](#), authored by D K Bhattacharyya, J K Kalita and S Roy; published by Elsevier, 2023.